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I SEMESTER, 2013-2014
COMPREHENSIVE EXAMINATION (OPEN BOOK)
(3 DEC. 2013) AN
EEE F443 ANALOG AND DIGITAL VLSI DESIGN

TIME - 3hrs.

MAX. MARKS=110

NOTE:

- Question paper has 4 pages, total 6 questions
- Write your ID No. , name, clearly on all the sheets.
- Answers should be clear, concise and legible. Specify your assumptions clearly.
- **Do all parts of same question together.** Diagrams and sketches should be neat and labeled. NO MARKS for unnecessary theoretical explanations
- Symbols have their usual notations. Read instructions/ parameter carefully
- → If not specified in question, ignore γ, λ wherever possible. Use square law for MOS transistor.
- → Wherever not shown, take bulk of NMOS connected to ground and bulk of PMOS connected to Vdd.
- → Answers without justification will not fetch any mark
- → Body connection is not shown. Neglect body effect unless asked to consider it.
- → Bode plots need not to be the scale, but must be qualitatively correct)
- If not given in the question, assume step input swinging from 0 to 3 V.
- Assume $\gamma=\lambda=0$ in drain current analysis unless necessarily required or specified in the question.

Unless specified in the question, assume bulk connected to ground for NMOS and bulk connected to Vdd for PMOS. Use square law for calculations unless mentioned specifically. Justify your answers

Unless Given use -- $V_{DD} = 3V$, $C_{ox} = 0.38pF/\mu m^2$, $kT/q = 26mV @ 300K$

For NMOS device---

$$\mu_n C_{ox} = 140 \mu A/V^2, V_T = 0.7 V, \lambda = 0.2 V^{-1}, L_{min} = 1\mu m, R_n = 13K\Omega / (W/L)_n,$$

For PMOS device---

$$\mu_p C_{ox} = 40 \mu A/V^2, V_T = -0.7 V, \lambda = 0.1 V^{-1}, L_{min} = 1\mu m, R_p = 31K\Omega / (W/L)_p,$$

Q 1 Consider fig. 1 and the Boolean expression given below----- assume input and their complements are available. Load capacitance at out/ out' is 20fF

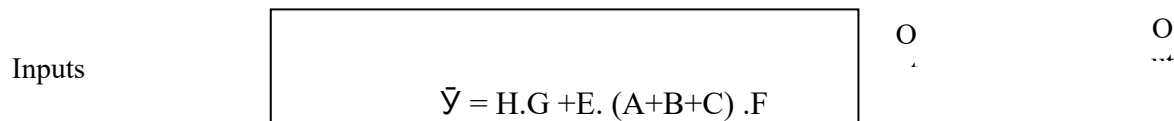


Fig. 1

- a) Realize the function $\bar{Y}$ using nmos pass transistor logic, sketch and label the worst case delay path
- b) Realize the function $\bar{Y}$ using domino logic. Sketch and label the worst case delay path. Assume clock signal is available
- c) For ABCDEFGH= 1011101, assuming intermediate node capacitive load is 1/2 of total capacitive load at out node of dynamic/ pass transistor network, determine the switching threshold voltage of static cmos inverter used before the output for charge sharing effect not to corrupt the output for both, the worst case delay, cases of part (a), and (b)
- d) Hence determine the (W/L) of transistors of static CMOS inverter in both cases of part (c)
- e) For design of part (a), it is desired to get output at rising edge of clock signal. Modify the design

of static cmos inverter to make it clocked CMOS inverter

- f) For design of part (b), sketch the transient waveform at out node, if each signal in input combination ABCEFGH=1011101 arrives after an interval of 1 nS from its predecessor. Hence, compute the total power consumption of entire design when out' node charges to V_{dd} . Neglect power consumption at input nodes.
- g) If clock signal to the domino CMOS gate has a routing length of 1mm, compute the wire delay required by the clock signal to reach the domino gate of part (b). Assume segment length of clock wire as 0.01 mm. take $(RC)_{\text{segment}} = 0.01\text{nS}$

25

Q 2 A two-stage, Miller-compensated CMOS op-amp is shown in figure 2. Capacitances C_1 and C_2 is the total capacitance at the output nodes of stage1 and stage2 respectively. (Ignore body effect)

- a) Derive the expression for the Transfer Function ($v_{\text{out}}(s) / v_{\text{in}}(s)$).
- b) Find expression for poles and zero's from the above transfer function. (Neglect the mirror pole). (Use the approximation that $C_2 \gg C_1$ and $C_c \gg C_1$)
- c) Write an expression for input common mode range and phase margin in terms of poles and zeros for part (ii).

Fig.

[15]

- d) Suppose Op-amp has a RHP zero at $20UGB$, a dominant pole (p_1) due to the Miller compensation, a second pole at p_2 and a mirror pole (p_3) at $-3UGB$.
 - (i) If UGB is 1MHz, find the location of p_2 corresponding to a 45° phase margin. Assume A_0 is very high.
 - (ii) Assume that in part (i) $|p_2| = 2UGB$ and a nulling resistor is used in series with C_c to cancel p_2 . What is the new phase margin assuming that $UGB = 1\text{MHz}$?
 - (iii) What is the phase margin if C_2 (load capacitance) is increased by a factor of 4 in part (ii)?
 - (iv) Physically, explain why the RHP zero occurs in the Miller Compensation.

[10]

Q 3 Consider a design which checks bitwise equality of 16-bit number, shown in Fig 3.

Assume, logical effort is 4 and parasitic effort is 4 P_{inv} for 2-input Ex-OR. Output load at Y[0] is 20 units, μ

$n / \mu_p = 2$, $P_{inv}=1$. Cap. Load at B[0] is 10 units

Fig.
3

For the path from B [0] to Y [0], Calculate-

- Path effort and stage effort.
- Minimum delay for given path.
- Capacitive load at each node of the given path.
- Justify if delay found in part (b) is minimum possible or not, If not, then redraw the path from B[0] to Y[0] for optimized design.

[20]

Q 4 Consider a digital System, which is implemented in a 1.2 V, 100nm process and has 200 million transistors, of which 20 million are in the logic gates and the remainder in memory arrays. The average logic transistor width is 12λ and the average memory transistor width is 4λ ($\lambda = 50\text{nm}$). The process has two threshold voltages. Subthreshold leakage for OFF devices is $20\text{nA}/\mu\text{m}$ for low threshold and $0.02\text{nA}/\mu\text{m}$ for high threshold devices. Memory use high threshold devices everywhere. Logic uses high threshold devices in all but 20% of the paths that are most critical for performance.

- If the subthreshold leakage is the primary source of the static power, Calculate the static power consumption of system.
- find out the static power consumption if only low threshold devices are available?
- What will be the effect on the speed performance of the system and justify it, if the logic is implemented in
 - all low threshold devices,
 - all high threshold devices.

[10]

Q 5 Figure 4 shows a CMOS inverter circuit with wire load capacitance, but with reduced input voltage swing. (Use: $\mu_n C_{ox} = 500 \mu\text{A}/\text{V}^2$, $\mu_p C_{ox} = 200 \mu\text{A}/\text{V}^2$ $V_T = \pm 1\text{V}$, $\lambda = 0$, $L_{min} = 1\mu\text{m}$)

- Compute the voltage levels, V_{OH} and V_{OL} for the given limits on V_{in} .
- Suppose V_{in} can swing rail-to-rail (0V to $V_{DD} = 5\text{V}$), then find t_{pLH} assuming an ideal step input by averaging the appropriate currents at the beginning and end of the output transition and assuming the capacitances as given in table 1 in addition to C_{wire} in the circuit.
- Now, assume that V

v_{in} can swing from (0V to 3.5V). Find the dynamic power consumption at the input and output for a given input switching frequency of 100 MHz and the capacitances as given in table 1 in addition to C_{wire} in the circuit. Comment and justify on value of static power consumption of the gate.

(15)

Tabl

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- Q 6 For the layout given in fig 5 and fig 6.
- In fig 5, Draw the equivalent labeled transistor level implementation. Also, calculate the size of the transistors
 - In fig 6, calculate value of the resistors. Take sheet resistance= 1.2 k ohms
- (15)